MorphMimic: Physics-Based Motion Imitation for Diverse Human Body Shapes

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Abstract

Animating digital humans with different body proportions requires references consistent with each body's geometry and a controller that tracks them in physics. We present MorphMimic: a motion corpus built from AMASS-derived HumanML3D clips, in which HUMOS generates shape-specific references for 128 body configurations (20,951 motion identities) that are paired with matching SMPL-based simulation assets, and one morphology-conditioned temporal controller trained on 18,855 of these identities across all bodies. MorphMimic achieves 99.14% tracking success on 1,048 held-out motions, with one selected trained body per clip. Replayed on 233 training-split clips across each of two panels of 128 unseen bodies, success is 97.28% inside the training coefficient range and 97.15% outside it, yet individual bodies succeed on as few as 3.0% and 13.3% of clips, and a later checkpoint improves outside-panel success while regressing individual bodies. In three-seed comparisons, attention variants reach higher mean success and lower position error than a temporal multilayer perceptron with less seed variability, while jerk differences stay within seed variation. An offline audit of 19,200 references shows that refinement reduces stance-foot sliding and below-floor body centres at the cost of higher reference jerk. The evidence supports strong average tracking and transfer within the evaluated domain; the causal contribution of explicit morphology inputs and the mechanism of the severe failures remain open.

Keywords: Physics-based animation; Digital humans; Motion imitation; Morphology-conditioned control; Temporal attention.

1 Introduction

Physics-based character animation should reproduce actions on digital humans with different proportions and dynamics, and a shared controller offers a way to reuse one motion repertoire across those bodies, provided that references and simulated assets stay consistent. Changes in limb lengths, mass distribution, and collision geometry alter both the reference trajectory and the dynamics required to follow it. Learning across body shapes requires a consistent relationship between source motion, shape-specific reference, and simulated body. Good tracking of new motions on a familiar body does not establish transfer to a new body, and a high average over bodies can conceal severe failures.

The building blocks exist: large motion resources, shape-conditioned reference generation, and scalable imitation [1, 2, 3, 5, 7]. Body-conditioned control is established [10], but the large-scale trackers Perpetual Humanoid Control (PHC) and Adversarial Skill Embeddings (ASE) learn from motion collections and report results on a single body shape; PHC supports body variation, although its quantitative evaluations use the mean body of the Skinned Multi-Person Linear model

(SMPL) [5, 6, 13] (Section 2). What remains open is how reliably one controller learns and transfers across a large corpus of explicitly paired motions and bodies.

We approach it with MorphMimic: a paired corpus in which every clip/body identity is preserved from HUMOS generation through simulation and replay (20,951 clip identities and 128 body configurations), and one shared temporal slot/type-attention controller that trains directly on 18,855 of those clips using streamed motion residency (Figure 1). Evaluation separates motion exposure from body exposure. The contributions are:

- the paired corpus and workflow: 20,951 clip identities on 128 body configurations (2,681,728 clip/body variants), each identity preserved from generation through simulation and replay;
- the shared controller with streamed residency, trained directly on 18,855 clips across all 128 bodies;
- exposure-separated evaluation of trained and unseen bodies: 1,048 clip-disjoint test clips on trained bodies, and 233 training-split clips replayed on 128 unseen inside-range and 128 unseen outside-range bodies (29,824 clip/body pairs per panel); and
- three-seed comparisons of four network architectures, and a reference-refinement audit of 19,200 development references (150 clips $\times$ 128 bodies).

We inherit the simulator integration, proximal policy optimization (PPO), and motion-tracking infrastructure from ProtoMotions and do not claim them, or morphology conditioning itself, as new. HumanML3D captions are metadata, not policy inputs: MorphMimic is reference-conditioned imitation, not text-conditioned control.

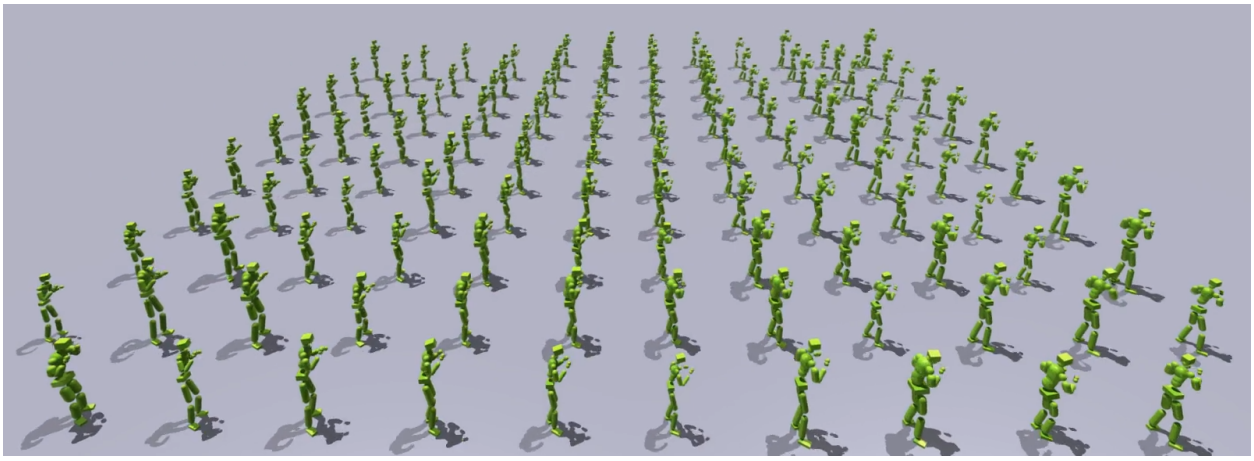


Figure 1: One motion tracked by MorphMimic on 128 trained bodies.

The original MorphMimic checkpoint achieves 99.14% success on 1,048 clip-disjoint test clips, each paired with one selected trained body. On 233 training-split clips across 128 unseen inside-range bodies and 128 unseen outside-range bodies, success is 97.28% and 97.15%, yet the worst bodies succeed on only 3.0% and 13.3% of those clips. These findings support the central claim: *one shared controller learns morphology-specific tracking with strong average transfer to unseen shapes, but uneven robustness across individual bodies*. The design evidence explains practical trade-offs; it does not yet isolate the benefit of explicit morphology inputs or establish uniform held-out-motion performance across all bodies.

The remainder of the paper is organized as follows. Section 2 reviews related work, and Section 3 gives an overview of the system and presents the corpus. Section 4 describes the policy and its training, Section 5 presents the evaluation, and Section 6 discusses limitations and concludes. The appendices provide reproducibility details, training provenance, and additional unseen-body results; the complete per-body tables are in Online Resource 1.

2 Related Work

Physics-based motion imitation. DeepMimic showed that reference-guided reinforcement learning can produce simulated character skills [4], and Perpetual Humanoid Control extended robust imitation to real-time simulated avatars [5]. Both motivate learning a feedback controller rather than replaying a kinematic trajectory. PULSE learns a motion representation for humanoid control, OmniH2O, ExBody2, and GMT study large-scale control on robot platforms [17, 18, 19, 20], and ASE learns reusable latent skills rather than exact clip tracking [6]; their published scores do not measure our clip/body replay task. We build on ProtoMotions with Isaac Gym and PPO [7, 8, 9].

Body variation and retargeting. Body-shape variation has been studied in physics-based control. Won and Lee learned a parametric controller over character height, weight, and proportions [10], and ShapeVerse trains characters with varied SMPL shape parameters to imitate walking and jogging [11]. Policies that transfer across changes in agent *topology*, such as limb count, use shared modular sub-policies [21], a different axis from the continuous variation studied here. Morphology conditioning is therefore established and not claimed as new; we examine one controller over a large set of paired motions and bodies, with shape identity maintained through generation, simulation, and evaluation. Kinematic retargeting and dynamic tracking meet different requirements. SMPL gives a common parameterized body [13], SMPLSim builds simulation-ready assets from it [14], and physics-based retargeting of motion capture to realistic body shapes has been studied with spacetime optimization [12]. A shared skeleton makes poses comparable but does not make global positions or required actuation independent of shape, so matching a reference to its simulation asset is part of the task definition.

Motion data and shape-conditioned generation. AMASS consolidates motion capture in a common body-model representation [1], and HumanML3D associates processed segments with natural-language descriptions [2]; we inherit these identities and descriptions rather than claim new capture or shape-specific annotation. HUMOS generates identity-conditioned motion with a cycle-consistent formulation [3], and we use its pretrained model to obtain shape-specific references, not as the controller. The sequences still need conversion to the simulator's skeleton and a feasibility check there, and our refinement is a deliberately limited kinematic post-process, not a new generative model or trajectory optimizer. The controller uses standard Transformer operations [15], and one ablation adapts the zero-initialized adaptive layer normalization (adaLN) conditioning of Diffusion Transformers [16] to actor-only morphology modulation; it tests a conditioning mechanism, not diffusion-based action generation.

Comparison with related systems. Table 1 lists the reported tracking results of PHC and ProtoMotions beside MorphMimic's. PHC, ASE, and ProtoMotions learn from motion collections but report results on a single body shape. PHC evaluates on the mean SMPL body, with optional shape-sampled training and no systematic unseen-body panel [5]; ASE reports task returns on

one sword-and-shield humanoid rather than tracking success [6]; and ProtoMotions documents configurable characters and a fall-based success rate [7]. Won and Lee establish parametric body control, with experiments that differ from our clip-by-body panels [10]; as neither they nor ASE report a comparable tracking success, both are left out of the table. PHC's success criterion matches ours (joint error above 0.5 m at any point), but motions, clip lengths, bodies, and reference sources (captured versus generated) differ, so the rows are not a ranking; no external system has been rerun under our protocol, our quantitative design comparison is internal, and no performance advantage is claimed. MorphMimic's distinction is the combination of many motions and many body shapes under one controller, with references and assets paired per body and systematic inside- and outside-range replay.

Table 1: Reported tracking results of PHC, ProtoMotions, and MorphMimic.

System	Training set	Motions	Evaluated on	Bodies	Succ. (%)	$E_{\text{g-mpjpe}}$ (mm)
PHC [5]	AMASS	11,313	AMASS-Train	1 (mean SMPL)	98.9	37.5
			AMASS-Test (140)	1 (mean SMPL)	96.4	47.4
ProtoMotions 3 [7]	AMASS	40+ h	AMASS (docs)	1 per character	≈99[a]	—
MorphMimic	HUMOS references from AMASS	18,855	Held-out test (1,048)	128 (one per clip)	99.14	67.6
			Inside range (233)	128 unseen	97.28	98.8
			Outside range (233)	128 unseen	97.15	95.1

Succ.: joint error $\leq$ 0.5 m at every frame (PHC's criterion; Eq. (12)). $E_{\text{g-mpjpe}}$: global mean per-joint position error, for MorphMimic over all clips including failures. [a]Fall-only criterion. "—": not reported. Both PHC and MorphMimic exclude object-interaction sequences.

Inherited functions and study additions. ProtoMotions provides the simulation abstraction, PPO training, and motion-library and imitation infrastructure, HUMOS supplies the pretrained generator, and SMPL and SMPLSim supply the body parameterization and assets. We add the paired corpus assembly and identity checks, the reference processing and fidelity audit, the evaluated slot/type controller configuration, streamed clip residency, and exposure-separated evaluation. These are system and empirical contributions built on existing methods, not claims to invent attention, PPO, body models, or motion imitation.

3 Overview

A body configuration $b = (g, \boldsymbol{\beta})$ specifies a SMPL template and shape coefficients. A *clip* denotes a source segment, including a mirrored segment when present, and a *variant* denotes that clip on one body configuration. For source clip c, HUMOS supplies a shape-specific reference $\hat{s}_{c,b}(t)$ and SMPLSim supplies the matching asset, so the corpus connects every clip identity to a motion description, 128 generated variants, and their matching simulation assets (Figure 2). Changing b changes geometry, reference positions, mass/inertia, and mass-scaled actuation even though topology is shared, so the task is to track these paired references with one policy rather than impose identical world-space paths on all bodies.

MorphMimic learns $\pi_\theta(a_t \mid o_t, b)$ from the present state, history, future reference targets, and body information, with the same parameters serving every body. Clips can change on reset while each environment retains its asset, and evaluation preserves reference/body identity. The flat-ground task has no external objects or partners, and captions are metadata, not controller input.

Motion	AMASS $\longrightarrow$ HumanML3D preprocessing and clip extraction $\longrightarrow$ HUMOS conditioned on $(g, \boldsymbol{\beta})$ $\longrightarrow$ MotionLib conversion and reference refinement $\longrightarrow$ per-clip motion variants.
Body	SMPL with the same $(g, \boldsymbol{\beta})$ $\longrightarrow$ SMPLSim $\longrightarrow$ matching skeleton, collision geometry, mass/inertia, and MuJoCo XML (MJCF) asset identity.
Text	HumanML3D source descriptions $\longrightarrow$ clip-ID lookup associated with the generated variants (metadata only; not a policy input).
The body assets are used for motion conversion, shape-specific forward kinematics, ground-clearance correction, and simulation. Refined reference motion and matching simulated state feed one shared policy.	

Figure 2: Paired reference and simulation-asset construction.

Source motion. We use the AMASS-derived motion portion of HumanML3D [1, 2], drawn from the 18 source collections listed in Appendix A.1. Source SMPL+H sequences are processed at 20 Hz, with clip extraction, temporal cropping, mirroring, and treadmill/skating cleanup following the HumanML3D/HUMOS pipeline, so HumanML3D identities are preserved. The pipeline produced 22,459 candidate clip identities, counting original and mirrored clips; a text-based filter removed 1,508 that appear to require external objects or environmental support, leaving 20,951 identities for flat-ground imitation. The filter is a task-scope heuristic and does not prove that every retained reference is support-free or dynamically feasible. HumanML3D descriptions stay associated with the variants of their source clip as metadata; their semantic fidelity is not audited (Appendix A.2), and they supply keyword-based motion classes only for the exploratory physical analysis in Appendix D.

Body configurations. A body configuration is $b = (g, \boldsymbol{\beta})$, where g identifies the female or male SMPL variant and $\boldsymbol{\beta} \in \mathbb{R}^{10}$ contains shape coefficients. We sample

$$\beta_{ij} \sim \mathcal{U}(-3, 3), \qquad i = 1, \ldots, 64, \quad j = 1, \ldots, 10, \tag{1}$$

and cross the 64 vectors with the two variants, giving 128 body configurations rather than 128 independent coefficient vectors. Uniform sampling is a controlled source of variation, not a representative sample of human populations.

Reference and asset generation. For each clip, the pretrained HUMOS model [3] receives the processed pose and root features and the target body parameters, and returns a shape-conditioned reference (root-relative 6D rotations with translation). In parallel, SMPL bodies [13] with identical parameters are converted to MJCF assets with SMPLSim [14], using mass estimation and hand and foot simplifications. Each asset has 24 rigid bodies and 69 actuated joint coordinates, with its own collision proxies, joint ranges, and estimated masses and inertias; these are simulation-model estimates, not measurements of human participants.

Reference refinement. Generated sequences are then converted to ProtoMotions' Z-up convention and resampled from 20 to 30 Hz. Forward kinematics (FK) uses the matching asset rather than a single neutral skeleton, and the packaged representation stores global body positions and rotations, local rotations, joint positions, velocities, and contact labels. Refinement is applied independently to each variant in five steps; default settings are in Appendix A.1. *Contacts:* ankle and toe contacts are validated, or detected from foot position and velocity when labels are unusable; short gaps are closed and very short runs removed. These label candidate stance intervals and are not measured forces. *Rotation filtering:* joint rotations are filtered with sign-continuous

quaternions, normalization, and a bounded geodesic correction, and contact transitions divide the sequence into segments so that filtering does not average across a change of stance. *Root correction:* a conservative least-squares root translation, anchored at each contact's median foot position, reduces horizontal stance sliding; it is temporally regularized and capped, so it does not guarantee exact foot locking and does not solve for new joint angles. *Ground clearance:* with floor height zero and margin ϵ_z, the lift required at frame t is

$$\ell_t^{\text{req}} = \max\left(0, \epsilon_z - \min_{x \in \mathcal{C}_b(t)} x_z\right), \tag{2}$$

where $\mathcal{C}_b(t)$ is the collision geometry of body b, and a smoothed lift envelope stays above this requirement. This prevents penetration of the checked geometry in the reference; it does not establish balanced support, realistic impacts, or freedom from self-collision. *Reconstruction:* FK and velocities are recomputed from the corrected root and rotations within each motion. The objective is modest, to reduce measurable kinematic artifacts while preserving the generated motion, and the refinement audit in Section 5 reports artifact and fidelity measures; improved kinematics alone does not demonstrate improved policy learning or dynamic feasibility.

Scale and splits. Each per-clip file stores the 128 variants with frame counts, sampling rates, and the clip and asset identities needed to recover body parameters, so the training system can stream clip files without downloading the complete corpus. Splits are by clip identity: all variants of a clip, and original/mirrored partners, share a partition (Table 2), and variants are generated combinations rather than independent captures. The split is versioned with seed 42 and roughly 90/5/5 proportions; the 150 development clips used for design comparisons and the reference audit ($150 \times 128 = 19{,}200$ variants) are reserved within the training split together with their mirrored partners, 286 identities in total, and the partners are not part of the 150-clip analysis (Appendix A.7); the test manifest is never loaded by the training workflow. The splits test held-out clip identities on the trained body set. They have not been shown to be disjoint by subject or capture recording, which limits claims about broader motion generalization.

Table 2: Corpus and split sizes.

Partition	Clip identities	Motion variants
Training	18,855	2,413,440
Validation	1,048	134,144
Test	1,048	134,144
Total	20,951	2,681,728

4 Policy

MorphMimic's controller is the policy $\pi_\theta(a_t \mid o_t, b)$ of Section 3: it maps the present state, history, future reference targets, and morphology to joint targets. This section defines its states and actions, network, reward, and training.

State. Experiments use the `smpl_mor` assets in Isaac Gym through ProtoMotions [8, 7], with simulation at 60 Hz, two substeps, and control decimation two, giving a 30 Hz policy rate. The

observation combines the present state, history, future reference targets, a historical action, and explicit morphology information. Let $B = 24$ denote the number of simulated bodies. The present state contains root height above ground, root-relative body positions excluding the root, 6D body rotations, and linear and angular velocities. Positions and velocities use a heading-aligned frame and rotations remove the root heading, which removes dependence on arbitrary global heading and horizontal origin while retaining height and body configuration. We add seven historical states at control-step offsets $\{1, 2, 3, 4, 8, 16, 32\}$, covering about 1.07 s and each in its own root/heading frame, and four future reference slots at $\{1, 2, 4, 8\}$, covering about 0.27 s. Future slots contain target positions relative to the current root, heading-aligned target rotations, position differences from the current body state, relative rotations $R_{t,j}^{-1}\hat{R}_{t+k,j}$ (expressed with respect to the current body rotation), and heading-aligned linear and angular velocity differences. The history exposes recent changes and the future targets a short anticipation window; contact flags, an explicit phase scalar, text, and estimated physical feature vectors are not part of the observation. Table 3 summarizes the inputs of actor and critic, with dimensions per slot, positions in metres, linear velocities in m/s, angular velocities in rad/s, and dimensionless 6D rotations.

Table 3: Observation inputs supplied to both actor and critic.

Input	Dimension	Slots	Representation and path
Current state	$15B - 2 = 358$	1	Height; $3(B-1)$ positions; $6B$ rotations; $3B$ linear and $3B$ angular velocities. Current-state encoder.
History	358	7	Same state representation at the seven past offsets. Shared history encoder.
Future reference	$24B = 576$	4	Target and relative poses ($18B$), plus velocity differences ($6B$). Shared future encoder.
Historical action	$D = 69$	1	A raw previous policy-action vector, before the tanh and joint-target mapping. Bypasses attention.
Morphology	11	1	$[g, \beta_1, \ldots, \beta_{10}]$ from the assigned asset. Bypasses attention.

Action. The policy outputs a raw action $a_t \in \mathbb{R}^D$ with $D = 69$, mapped to joint targets by

$$q_t^{\text{target}} = q^{\text{offset}} + q^{\text{scale}} \odot \tanh(a_t), \tag{3}$$

with offsets and scales derived from the configured joint ranges. The targets drive the simulator's built-in position proportional–derivative (PD) controller; the action is not a residual added to the reference pose. PD stiffness, damping, and effort limits are scaled per joint by the mass of the associated rigid body relative to that body in the reference asset (base values in Appendix A.4). This body-level mass scaling is not a single scale based on total character mass, and it does not guarantee dynamically equivalent actuation across geometrically different bodies.

Network. MorphMimic uses temporal slot/type attention. Three observation encoders, one for the current state, one shared across history slots, and one shared across future slots, are multilayer perceptrons (MLPs) with two 256-unit hidden layers, rectified linear unit (ReLU) activations, and a 256-dimensional token output, and their inputs use running mean/standard-deviation normalization with a magnitude clamp of five. The resulting 12 tokens (one current, seven historical, four future) enter a Transformer as

$$z_i = E_{\tau(i)}(x_i) + e_i^{\text{slot}} + e_{\tau(i)}^{\text{type}}, \qquad i = 0, \ldots, 11, \tag{4}$$

where $\tau(i)$ identifies the current, history, or future source: slot embeddings distinguish the configured temporal offsets, and type embeddings the observation sources. The Transformer has two layers, four attention heads, token width 256, and feed-forward width 1,024 [15], and its current-state token supplies the pooled output after the configured output projection and activation. Figure 3 traces the encoded observations through attention and the separate actor and critic heads, which share input categories but not parameters.

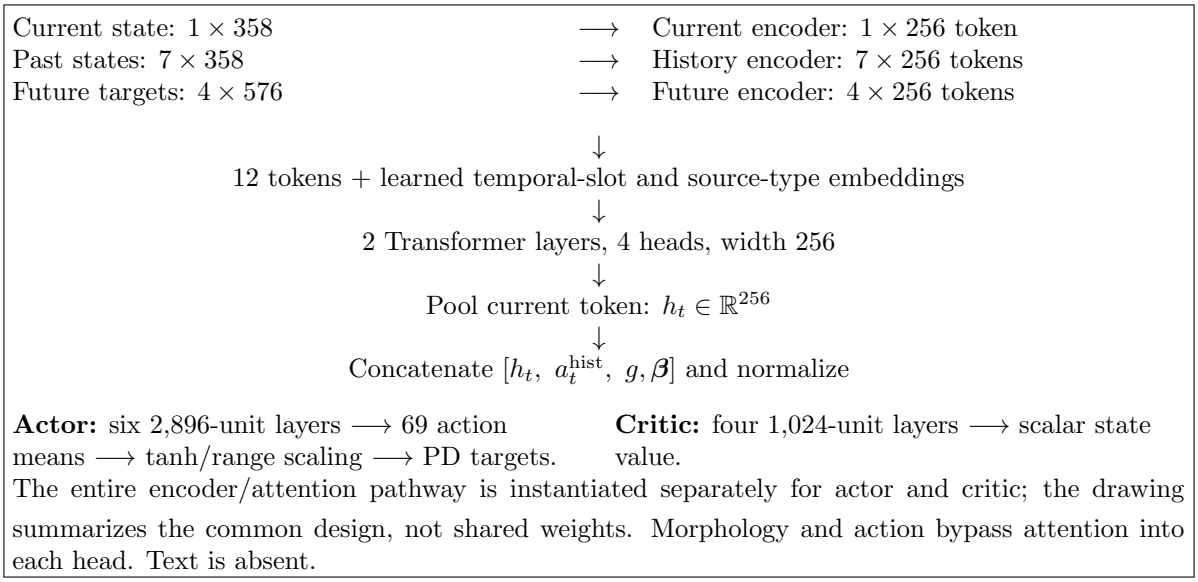


Figure 3: Temporal slot/type controller and separate actor/critic heads.

The final head input concatenates the 256-dimensional attention output, the historical raw action, and the 11-dimensional morphology vector; this 336-dimensional input also uses running normalization and clamp five. Betas are not manually divided by three, embedded as a separate attention token, or used for adaLN modulation in the selected model. Actor and critic have separate token encoders, attention modules, slot/type embeddings, and MLP heads, with six hidden layers of width 2,896 for the actor and four of width 1,024 for the critic, all with ReLU activations; they output a Gaussian action mean and a scalar value, respectively. PPO samples actions with fixed diagonal log standard deviation -2.9, and evaluation uses the mean action.

Reward. The training reward combines global body tracking and root-height tracking. For corresponding simulated and reference body positions $p_j, \hat{p}_j$, rotations $R_j, \hat{R}_j$, and velocities $v_j, \hat{v}_j$ and

$\omega_j, \hat{\omega}_j$, the errors are defined as follows:

$$e_p = \frac{1}{3B} \sum_{j=1}^{B} \|p_j - \hat{p}_j\|_2^2, \tag{5}$$

$$e_R = \frac{1}{B} \sum_{j=1}^{B} d_{\mathrm{SO}(3)}(R_j, \hat{R}_j)^2, \tag{6}$$

$$e_v = \frac{1}{3B} \sum_{j=1}^{B} \|v_j - \hat{v}_j\|_2^2, \tag{7}$$

$$e_\omega = \frac{1}{3B} \sum_{j=1}^{B} \|\omega_j - \hat{\omega}_j\|_2^2, \tag{8}$$

$$e_h = (h - \hat{h})^2, \tag{9}$$

Above, $d_{\mathrm{SO}(3)}$ is the geodesic rotation angle in radians, and the coordinate-wise means in the position and velocity errors match the implemented reductions. The reward is

$$\begin{aligned} r_t = {} & 0.5 \exp(-25 e_p) + 0.3 \exp(-5 e_R) + 0.1 \exp(-0.5 e_v) \\ & + 0.2 \exp(-0.1 e_\omega) + 0.2 \exp(-100 e_h), \end{aligned} \tag{10}$$

with weights that are not normalized to sum to one. There is no explicit effort, action-smoothness, or contact-matching reward in this experiment, and no source-dependent switch between canonical and HUMOS reward functions.

Optimization. MorphMimic is trained directly across all bodies on refined HUMOS references, with no neutral pretraining stage, using PPO [9]. References are sampled only from variants matching the environment's asset. Reference-state initialization starts at frame zero with probability 0.2 and otherwise at a random valid reference time. Training episodes last at most 1,000 control steps and terminate on falls (a 0.15 m height setting, with permitted foot-contact bodies), while tracking-error termination is disabled, so this rule is distinct from the evaluation failure threshold of Section 5. PPO uses 32-step rollouts, discount 0.99, generalized advantage estimation (GAE) parameter 0.95, policy-ratio clipping 0.2, normalized advantages, and separate Adam optimizers with learning rates 2×10^{-5} for the actor and 10^{-4} for the critic. The reported full-scale run uses training seed 0. Other settings and the actual run budgets are in Appendix A.3.

Streamed motion sampling. The full-scale experiment streams per-clip files through `GlobalClipPool`. The training manifest is deterministically partitioned across GPU ranks, and each rank keeps a bounded resident pool containing all 128 variants of its selected clips, with environment-level motion sampling respecting the assigned morphology. The pool is rebuilt every 256 epochs: most slots are selected by priority, combining estimated clip difficulty and an exploration bonus, and a fraction 0.2 is reserved for uniform random selection to maintain rehearsal outside the highest-priority set. This is a training curriculum, not uniform sampling of the complete corpus. Validation membership is fixed independently of pool rebuilding and does not update sampling weights, so resident-pool and validation metrics measure different populations. Sampling constants and the pool size actually used are in Appendices A and B.

5 Evaluation

We evaluate MorphMimic on held-out motions, on trained bodies, and on unseen bodies, and use per-body failures to delimit the claims. Each reference is rolled out only in an environment using its exact asset identity, so a reference is never assigned to a different body. Evaluation uses the mean action and reference initialization, disables training termination, and records errors over the valid reference horizon, capped at 600 control steps (20 s at 30 Hz), so success refers to the evaluated horizon, not necessarily the whole source clip. Held-out evaluations select one body variant per clip with a deterministic hash of the clip identity (Appendix A), so the clip list is fixed while the body panel rotates across scheduled evaluations. Validation was monitored repeatedly and is not an untouched test set, whereas the clip-disjoint test manifest stays outside training and model selection and is evaluated once.

Design comparisons run on a development corpus of 150 clips and all 128 body configurations (19,200 variants), reserved within the training split; it is not a held-out generalization benchmark. Because compute was limited, architecture and reference-processing choices are investigated on this subset with a fixed simulation budget, whereas the full-scale run trains on the 18,855 training clips (Table 2). An exploratory analysis of cross-body control demand on the same clips was inconclusive and is reported in Appendix D. Table 4 lists all populations; held-out identity is clip-grouped, not a demonstrated subject or recording split.

Table 4: Evaluation populations; each unseen panel has 128 bodies.

Evaluation	Motion exposure	Body exposure	Pairs
Validation / test	Held-out clips	Trained; one per clip	1,048 each
Development replay	150 training clips	All 128 trained	19,200
Unseen inside / outside	233 training clips	All 128 unseen per panel	29,824 each
Unseen diagnostic subset	23 val/test clips; seen by full-data	Same unseen panels	2,944 each
Trained-body control	150 training clips	All 128 trained	19,200
Design comparison	Development clips	Trained bodies	Three refined seeds

Metrics. For each evaluated clip/shape pair i, the per-frame mean body position error is

$$d_{i,t} = \frac{1}{B} \sum_{j=1}^{B} \|p_{i,t,j} - \hat{p}_{i,t,j}\|_2, \tag{11}$$

and the configured success indicator is

$$S_i = \mathbf{1}\left[\max_{t<T_i} d_{i,t} \leq 0.5 \text{ m}\right], \tag{12}$$

where T_i is the evaluated horizon. This threshold is deliberately distinct from the smoother training reward, and passing it does not imply fine-grained pose accuracy or biomechanical realism. We also report success at the stricter thresholds 0.3, 0.2, and 0.1 m, time-averaged mean body-position error (the global mean per-joint position error $E_{\text{g-mpjpe}}$ of Table 1), mean global body-rotation geodesic error (not a local joint-angle error), the time average of the largest body-position error within each frame (the logged *mean maximum-joint error*), and the windowed normalized jerk of Appendix A.5, whose high-jerk statistic is the percentage of windows in which any body exceeds 6,500. Errors are averaged over valid frames per motion and then over motions, including failed rollouts; offline comparisons aggregate saved per-motion records over unique clip/shape pairs.

Table 5: Offline evaluation of MorphMimic (epoch 28,170) on validation and test splits.

Metric	Validation	Test
Success at 0.5 m (%) ↑	98.47	99.14
Nominal 95% Wilson interval, 0.5 m (%)	[97.53, 99.06]	[98.38, 99.55]
Success at 0.3 m (%) ↑	95.52	96.28
Success at 0.2 m (%) ↑	84.83	85.97
Success at 0.1 m (%) ↑	37.88	38.55
Failure rate at 0.5 m (%) ↓	1.53	0.86
Mean body-position error (m) ↓	0.0699	0.0676
Mean global body-rotation error (rad) ↓	0.1738	0.1723
Mean maximum-joint position error (m) ↓	0.1419	0.1393
Normalized jerk ↓	1,154.3	1,050.9
High-jerk windows (%) ↓	27.46	25.33
Worst single clip, gt err max (m)	3.79	1.67

Held-out motions on selected trained bodies. The epoch-28,170 MorphMimic checkpoint was re-evaluated offline with a standalone evaluator, once on the validation split and once on the clip-disjoint test split, with one selected body per clip (shape panel index 110; Table 5). All 1,048 clips of each split last about 6.6 s (198 evaluated frames), so none reaches the 600-step cap. The test manifest is loaded only by this evaluator, which first verifies its hash, clip count, and disjointness from train and validation; the test split was evaluated once and not used to select the checkpoint.

Validation and test scores are similar (1,032 and 1,039 of 1,048 successes at 0.5 m), and no significance claim is made from their difference. The 0.5 m threshold is permissive, so the table also reports stricter thresholds and continuous errors: test success falls to 96.28% at 0.3 m, 85.97% at 0.2 m, and 38.55% at 0.1 m. Tight thresholds partly measure the references rather than the controller, because HUMOS generates them kinematically and they are not dynamically validated; the refinement audit below finds residual contact artifacts even after refinement. The Wilson intervals [22] are nominal: they condition on the checkpoint and selected body panel and ignore dependence between related recordings or mirrored clips, training variability, and uncertainty across body selections.

Coverage of trained bodies. Replaying 150 training clips on all 128 trained bodies (19,200 pairs) reaches 99.29% success with the same checkpoint (Table 6), which establishes broad tracking on that subset. All-body consistency on held-out motions remains unmeasured: it would need a held-out-motion panel crossed with all 128 bodies. Figure 1 shows one motion tracked on the 128 trained bodies; it illustrates execution and does not quantify tracking, contact quality, or transfer.

Residual failures. Nine test and 16 validation pairs fail the threshold. Their captions suggest ground transitions (six test, five validation failures) and whole-body rotations (two test, four validation), and also mention turning, side-stepping, an abrupt stop, and a hand-held sparring stick absent from simulation. These describe the failed sample rather than mechanical causes or class-specific failure rates; diagnosis would need synchronized rollouts and first-crossing traces, including successful variants of the same clips.

Unseen-body panels. Each panel contains 64 new coefficient vectors instantiated on male and female SMPL templates. Inside vectors lie in $[-3, 3]^{10}$; outside vectors have $\|\beta\|_\infty \in (3, 5]$, with 16 vectors (32 bodies) in each of four half-unit tiers. Inside denotes membership of the training

coefficient box, not of the convex hull of trained bodies. Both panels use matching HUMOS references and assets, frame-zero offsets, and full reference refinement; the build report records zero penetration under its refinement diagnostic but does not certify dynamic feasibility. Replay uses mean actions, reference initialization, 32 environments per asset, and at most 210 control steps (7 s), with success as in Eq. (12). The 233 training-split identities isolate the body axis, while the other 23 validation/test identities remain separate; the trained-body control uses a different 150-clip panel and refinement run, so only within-dataset checkpoint deltas are paired comparisons.

High average transfer inside and outside the training range. Crossing the 233 training-split clips with all 128 bodies in each panel gives 29,824 pairs per panel, and Table 6 includes every body and every failed pair. MorphMimic's original checkpoint reaches 97.28% inside-range and 97.15% outside-range success. The trained-body control reaches 99.29%, but uses different clips and a different refinement run, so the gap of roughly two percentage points (pp) is descriptive, not a matched estimate of the effect of changing bodies. Deltas are paired within each dataset, 95% confidence intervals (CI) resample clips with the body panel fixed, and the rerun repeats the original checkpoint, not its training.

Table 6: Unseen-body success of the original and later checkpoints, with the trained-body control.

Panel	Refined (%)	Full-data (%)	Rerun Δ (pp)	Candidate Δ [95% CI] (pp)
Inside unseen	97.28	96.91	−0.02	−0.37 [−0.57, −0.18]
Outside unseen	97.15	97.57	+0.04	+0.42 [+0.22, +0.61]
Trained control	99.29	99.12	+0.02	−0.17 [−0.37, −0.03]

Failures concentrate in individual bodies. High averages coexist with severe body-specific failures. Inside body `male_669a0fb3` succeeds on 3.0% of clips and outside body `male_1c099619` on 13.3%; another inside body, `male_ec3bb7f8`, reaches only 19.3% despite an in-range mass of 42.7 kg and height of 1.41 m. Figure 4 shows all 128 bodies of each panel, sorted independently by original-checkpoint success on the same 233 known clips (the ranking does not pair inside and outside bodies), and Online Resource 1 reports every body. Lying outside the training coefficient box does not explain these failures: success does not decline monotonically across the four outside coefficient tiers (99.03, 97.76, 95.33, and 96.47%). Bodies outside the trained height range perform well on average (97.91% outside; 96.42% inside), while the tallest in-range bin is weaker (90.53% outside; 88.75% inside). The eight heaviest outside bodies, above 144.4 kg, average 79.02%, but small overlapping property groups and paired male/female templates are not independent samples of physical-property space, so coefficient extremity, mass, and height give no simple ordering of transfer difficulty (tier and mass tables in Appendix C). All failed rows have zero recorded termination flags; because flags depend on the replay configuration rather than an independent fall detector, we treat these as threshold failures pending synchronized rollouts and contact and root-height traces.

Checkpoint sensitivity. The later full-data checkpoint (epoch 34,000) was trained on all original motion splits and differs from the original (epoch 28,170, distinct from the epoch-28,159 monitoring snapshot of Appendix B) by −0.37 pp inside and +0.42 pp outside (Table 6). Paired 95% bootstrap intervals resample clips with the body panel fixed, not training seeds or unseen bodies, and a change counts as beyond noise only if its interval excludes zero and its magnitude exceeds the identical-checkpoint rerun delta; both changes qualify, but duration and motion coverage changed together, so

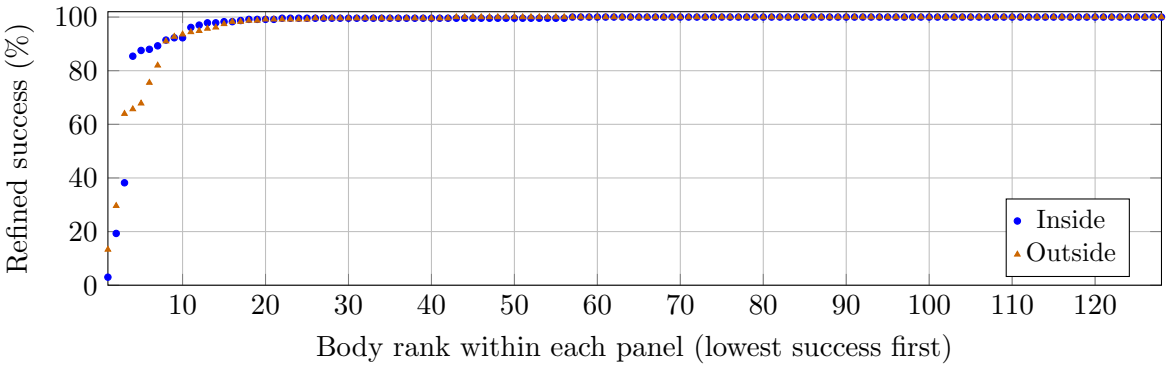


Figure 4: Unseen-body success, sorted within each panel.

they cannot be attributed to either, and without a checkpoint ladder ordinary checkpoint variation remains unmeasured. Identical-checkpoint reruns agree on 99.33% of inside and 98.68% of outside pair labels (99.96% in the control), yet individual pairs can change by metres in peak error, so small pooled changes coexist with local instability. The outside gain is concentrated in the ten bodies beyond the trained mass range (+7.68 pp, 95% CI [+6.39, +9.01]) and the outermost coefficient tier (+1.39 pp [+1.07, +1.72]), while mean position error rises by 2.4, 5.0, and 3.1 mm in the outside, inside, and control panels, so threshold success and continuous tracking move differently. Three inside and six outside bodies regress by more than 2 pp with intervals excluding zero, by up to 36.5 pp (per-body intervals are exploratory and unadjusted for multiple testing), and five outside bodies improve by more than 2 pp, including `male_1c099619` (13.3% to 71.2%) (Appendix C); neither improved extremes nor a higher outside average justify calling the later checkpoint uniformly better. On the 23-clip diagnostic subset, which the later checkpoint trained on, inside success is 97.08% versus 96.50% and outside 96.81% versus 96.91% for the original and later checkpoints; these values are excluded from the headline. Severe failures remain unresolved: per-asset reference feasibility and synchronized contact, root-height, and actuation traces have not been audited, so reference, controller, and asset mechanisms cannot be distinguished.

Design comparisons. We compare four architectures: A, temporal MLP; B, temporal attention; C, attention with slot/type embeddings, the architecture of MorphMimic; and D, C with actor-only adaLN-Zero. All retain morphology inputs in actor and critic. Refined-reference runs use three seeds (0, 1, 2) and a matched 6,000-epoch budget on the development subset (Table 7), and each run contributes one summary, the mean of its last eight evaluations (epochs 4,600–6,000); means and sample standard deviations are computed across the three run summaries, and within-run evaluations are not replicates. Seed-1/2 values were checked against local TensorBoard records, whereas the original logs of the seed-0 values were not independently available. The full-scale MorphMimic model predates these reruns, which reassess its design rather than establish prospective selection. Cases E–H repeat A–D with unrefined references and one seed each (Appendix B).

Position and rotation errors are across-seed means, and parameters are actor and critic totals. Attention variants reach 97.2–97.9% mean success against 95.1% for the temporal MLP, with lower position error (0.083–0.090 versus 0.118 m) and much smaller seed variability (standard deviation 0.08–0.34 versus 2.07 points), although the MLP has the most parameters, roughly 26–29% more than the attention variants. Among the attention variants, C has the highest mean success and D the lowest position and rotation errors. A has the lowest mean jerk, but jerk differences between architectures are comparable to the seed-to-seed standard deviations (643–887), so the apparent

Table 7: Three-seed refined-reference comparison, mean ± standard deviation across seeds.

Architecture	Success (%)	Pos. (m)	Rot. (rad)	Jerk	Params (M)
A: temporal MLP	95.14 ± 2.07	0.118	0.216	$2{,}110 \pm 643$	65.9
B: temporal attention	97.56 ± 0.34	0.090	0.218	$2{,}783 \pm 887$	51.2
C: slot/type attention	97.89 ± 0.22	0.090	0.219	$2{,}413 \pm 762$	51.2
D: C + adaLN-Zero	97.17 ± 0.08	0.083	0.211	$2{,}659 \pm 674$	52.1

factor-of-two jerk advantage in seed 0 is not a consistent architecture effect. Three seeds do not establish statistical superiority, and MorphMimic uses C as a practical compromise.

Reference refinement. A paired offline audit of all 19,200 development references (150 clips × 128 bodies; no simulation) compares artifact and fidelity measures before and after refinement (Table 8). Artifact rows give the corpus mean before and after and the fraction of motions that improve; fidelity rows give the corpus mean refined-versus-unrefined difference. Refinement lowers stance-foot speed during detected contact from 0.055 to 0.038 m/s, with 96% of references improving, and below-floor frames from 39.6% to 3.3% (only 45% of references improve on the latter because many were already grounded). Measured motion features are largely preserved: the mean per-body rotation change is 0.12°, root horizontal displacement is 0.008 m, and the cross-shape variance of knee-flexion range is retained at a ratio of 0.94. The cost is higher reference jerk (the raw mean third difference of body positions, not the windowed evaluator score), from 45.2 to 60.7 $\mathrm{m/s^3}$, with 79% of references worsening; temporal root and clearance corrections are a candidate explanation, but their contribution to policy smoothness is not isolated. Refined and unrefined policies (cases A–D versus E–H) track different targets, so their comparison describes different tasks and does not establish a causal benefit of refinement, which would need a common-reference checkpoint comparison.

Table 8: Reference-refinement audit over 19,200 development references.

	Unref. mean	Refined mean	Motions improved
Stance-foot speed (m/s) ↓	0.055	0.038	96%
Frames below floor (%) ↓	39.6	3.3	45%
Frames within 2 cm of floor (%) ↓	49.8	33.6	—
Position jerk ($\mathrm{m/s^3}$)	45.2	60.7	21%
Per-body rotation change (deg)		0.12	
Root horizontal displacement (m)		0.008	
Cross-shape var., knee-flexion range (ratio r/u)		0.94	

Scope of the morphology-conditioning evidence. Every architecture receives morphology inputs in actor and critic. The adaLN-Zero variant changes how the actor uses them but does not test their removal, and body-dependent state and reference targets also expose morphology. Successful transfer therefore establishes a capability of the whole system, not that explicit coefficients cause it. Attributing it to those inputs needs a matched training run without them, with the same references, body panel, budget, and seeds; inference-time masking or shuffling would introduce distribution shift and cannot replace that comparison.

6 Discussion and Limitations

MorphMimic combines paired morphology-specific references and assets with one temporal controller and an evaluation that separates motion exposure from body exposure. Trained across 128 bodies, it achieves 99.14% success on held-out motions with one selected trained body per clip and executes development clips across all trained bodies. It transfers to unseen bodies at 97.28% inside and 97.15% outside the training coefficient range, but individual bodies can fail on most clips, and a later checkpoint introduces both recoveries and regressions. Temporal attention improves tracking accuracy and seed stability over a temporal MLP without a consistent smoothness difference, and the reference audit shows reduced contact artifacts at the cost of higher reference jerk. These comparisons support practical architecture and reference-processing choices, but not a causal explanation of transfer.

Implications for character animation. Body-specific references and one shared feedback controller support reuse across the tested digital-human configurations. The reference audit and the simulated-motion results are complementary: reference cleaning does not establish lower sliding or penetration in controller outputs, which remain unassessed, and high threshold success is tracking capability, not a complete assessment of visual realism.

Average transfer and individual robustness. Similar inside and outside averages do not make body extrapolation uniformly reliable. Success is non-monotonic across coefficient tiers, and poor tracking occurs both beyond and within the training mass and height ranges; these associations describe the tested assets and do not identify failure causes. The later checkpoint recovers some bodies and harms others, so threshold success alone would overstate its improvement.

Coverage and inference. The unseen-body headline uses known motion identities to isolate body transfer, the 1,048-clip test uses one trained body per clip, and a fixed held-out-motion panel crossed with all 128 trained bodies remains missing. Original and mirrored variants stay together, but the splits are not shown to be subject- or recording-disjoint. Clip-bootstrap intervals hold the body panel fixed and do not quantify uncertainty over shape space, and one full-scale run with one candidate checkpoint does not measure training-seed or ordinary checkpoint variation. Compared with PHC's reported results on the mean body (Table 1), MorphMimic's success rate is similar but its global position error is higher; the rows differ in motions, clip lengths, bodies, and reference source (captured versus generated), so we draw no ranking.

Mechanism and design. Attributing transfer to explicit morphology inputs needs the matched run described with the design comparisons (Section 5); the architecture comparison keeps those inputs in every model and has unequal capacities, with the temporal MLP the largest. Refinement reduces contact artifacts with small pose changes but raises target jerk, and policies evaluated against different targets do not isolate its learning benefit. The exploratory physical analysis (Appendix D) remains descriptive.

Failure diagnosis and scope. Failures are error-threshold failures with zero termination flags, which alone do not prove the absence of falls. Synchronized rollouts, contact and root trajectories, and per-asset feasibility audits are needed before assigning a mechanism; extreme HUMOS references have been refined but lack such validation. Estimated masses, collision proxies, and

mass-scaled gains are modeling choices, and tracking success and jerk do not certify biomechanical realism. The short, reference-initialized flat-ground protocol excludes objects, partners, long-horizon recovery, skeletal-topology changes, and real robots. The corpus is derived, so clip/body variants are not independent captures, and caption alignment is not audited. A matched external tracker would strengthen comparative conclusions, but the published capability comparison alone cannot establish a performance advantage.

Summary and outlook. The evidence supports strong average shared tracking and transfer within this protocol. All-body held-out-motion coverage, matched conditioning and external controls, and dynamic diagnosis of severe failures remain necessary for stronger claims about robustness and mechanism.

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Author contributions Longzhen Han conceived the study, developed the methodology and software, generated the data, and ran most of the experiments and analysis. Beck Farrington-Wheeler contributed substantially to the training experiments and the non-software work, with further contributions from Sajjad Hussain and Awes Mubarak. Almas Baimagambetov and Nikolaos Polatidis supervised the work. All authors approved the final manuscript.

Data and code availability The motion corpus derives from AMASS and HumanML3D and remains subject to the source datasets' access and licensing conditions. The research code repository is https://github.com/hansen1416/morph_mimic; demonstrations are available at https://hansen1416.github.io/morph_mimic/. Evaluation summaries and per-body results used in this manuscript are retained with the research records. [Authors to specify the archival code version, evaluated checkpoints and assets, evaluation-data access location, and any restrictions on derived data.]

AI assistance in manuscript preparation An AI assistant was used to help restructure and draft manuscript text and check consistency with the recorded sources. Reported experimental measurements come from existing research records; this assistance did not produce new training or rollout results.

A Reproducibility Details

A.1 Source preparation and refinement settings

The AMASS source collections used by the preparation pipeline are ACCAD, BioMotionLab_NTroje, BMLhandball, BMLmovi, CMU, DFaust_67, EKUT, Eyes_Japan_Dataset, HumanEva, KIT, MPI_HDM05, MPI_Limits, MPI_mosh, SFU, SSM_synced, TCD_handMocap, TotalCapture, and Transitions_mocap. Source preparation operates at 20 Hz; the final motion libraries use 30 Hz. Joint ordering and quaternion conventions must follow the matching asset and MotionLib conversion rather than be inferred from source array positions. Body coefficients are sampled with NumPy's random number generator using seed 46, and bodies are identified by stable beta keys rather than content hashes. SMPLSim assets use mass estimation and hand and foot simplifications (`real_weight`, `replace_feet`, `remove_toe`, `freeze_hand`). Table 9 lists the default reference-refinement settings.

Table 9: Default reference-refinement settings (frame counts at 30 Hz).

Operation	Setting
Contact candidates	Left/right ankle and toe bodies
Contact detection thresholds	Speed 0.15 m/s; height 0.10 m
Contact cleanup	Close gaps up to 2 frames; remove runs shorter than 3 frames
Rotation filter	Savitzky–Golay, 9-frame window, polynomial order 3
Rotation blend / bound	Strength 0.5; maximum geodesic correction 8°
Contact-transition guard	Separate segments; 2-frame transition taper
Root XY correction	Median stance anchors; least-squares common translation
Root correction regularization	21-frame smoothing window; 0.10 m displacement cap
Ground clearance	0.005 m collision-geometry margin; 5-frame lift smoothing
Final reconstruction	Per-asset FK and per-motion velocity recomputation

A.2 Caption provenance

HumanML3D descriptions are available through a separate clip-ID lookup and the annotation metadata used by HUMOS. A description is associated with the variants of its source clip; it is not a new annotation for each generated body. A coarse check against a single-caption-per-clip lookup finds a caption for all 20,951 clip identities, but semantic fidelity is not established: of the 9,579 mirrored identities, 22% carry a caption byte-identical to their un-mirrored source and 40% contain an explicit left/right or clockwise term that mirroring inverts, so removing a mirror prefix does not establish a correct caption. Segment timing and multi-annotation coverage against the full HumanML3D annotation set are not audited, and the keyword-based motion classes used in the physical analysis (Appendix D) have not been manually validated. The retained and excluded ID lists provide the audit trail for the object-support filter.

A.3 Training configuration

Table 10 records the shared optimization settings and the environment and batch sizes actually used; run-specific resolved configurations take precedence over older launch examples. The full-scale frame count is a recorded stopping snapshot, not a budget matched to the development runs. Clip priority combines estimated difficulty, updated from training-pool evaluations with an exponential

moving average of rate 0.1 and a floor of 0.05, and an exploration coefficient of 1.0; the pool is rebuilt every 256 epochs and a fraction 0.2 of slots is reserved for uniform selection.

Table 10: Training configuration; environment and minibatch counts are per GPU.

Setting	Corrected 150-clip runs	Full-scale MorphMimic run
Parallel GPUs per policy	1	6
Environments per GPU	4,096	2,048
PPO minibatch size per GPU	16,384	8,192
Rollout length	32 control steps	32 control steps
Mini-epochs per update	1	1
Actor / critic learning rates	2×10^{-5} / 10^{-4}	2×10^{-5} / 10^{-4}
Discount / GAE / ratio clip	0.99 / 0.95 / 0.2	0.99 / 0.95 / 0.2
Gradient norm clip	50	50
Evaluation cadence	200 epochs	256 epochs
Training seed	0–2 (refined A–D); 0 (unrefined E–H)	0 (saved run configuration)
Reference residency	All 150 clips	256, later 128 clips per GPU
Training frames	786,432,000 per run (6,000 epochs)	Approximately 11.07×10^9 recorded
Status of reported evidence	Completed, Table 7	Epoch-28,170 checkpoint, Table 5

A.4 PD gains and limits

Table 11 gives the base values configured for the SMPL morphology robot. For joint d belonging to rigid body j, the simulator applies the multiplier

$$\rho_{b,j} = \frac{m_{b,j}}{m_{\mathrm{ref},j}} \tag{13}$$

to stiffness, damping, and effort limit. The velocity limit is not mass-scaled. All relevant asset masses, joint ranges, collision settings, and the reference asset must accompany a reproducible physical interpretation.

Table 11: Base PD gains and limits before per-body mass scaling.

Joint family	Stiffness (Nm/rad)	Damping (Nm s/rad)	Effort limit (Nm)	Velocity limit (rad/s)
Hip, knee, ankle	800	80	500	100
Toe	500	50	500	100
Torso, spine, chest	1,000	100	500	100
Neck, head, thorax, shoulder, elbow	500	50	500	100
Wrist, hand	300	30	500	100

A.5 Windowed jerk convention

The logged smoothness score is computed from simulated global body positions using consecutive finite differences with control interval $\Delta t = 1/30\,\mathrm{s}$. A nominal 0.4 s window contains $W = \max(4, \lfloor 0.4/\Delta t \rfloor)$ frames, giving $W = 12$ at the configured rate and duration $T_w = (W-1)\Delta t$. For body j in window w, let $L_{w,j}$ be the sum of speed times Δt and let $\dddot{p}_{w,j}$ denote the third finite

difference. The implementation uses

$$J_{w,j} = \frac{T_w^5 \sum_{t \in w} \| \dddot{p}_{t,j} \|_2^2 \, \Delta t}{\max(L_{w,j}, \epsilon)^2 + \epsilon}, \qquad \epsilon = 0.1, \tag{14}$$

where the sum includes the valid third-difference samples in the window. Window scores are averaged per body and then across bodies and motions. High-jerk percentage counts windows for which $\max_j J_{w,j} > 6{,}500$.

Equation (14) documents the existing numerical convention, including its regularizers; it is not presented as a universally dimensionless or sampling-rate-invariant biomechanical measure. Comparisons must keep the sample rate, window, units, regularizers, and aggregation fixed. Reference refinement reports also include raw jerk in $\mathrm{m/s^3}$; those values are a different metric and cannot be compared numerically to this logged score.

A.6 Evaluation panel selection

Routine monitoring selects one variant per clip. With seed $s = 42$, stable clip identity c, panel index p, and N_c available variants, the choice is

$$k(c, p) = \left(\mathrm{uint64}_{\mathrm{big}}\left(\mathrm{SHA256}(s : c)_{0:8} \right) + p \right) \bmod N_c. \tag{15}$$

Here $s : c$ is the UTF-8 string `seed:clip_id`. For scheduled monitoring, p is the integer quotient of the checkpointed epoch and the evaluation cadence, which gives deterministic staggered shape choices that rotate across evaluations and remain reproducible after resume. The validation clip list is fixed while the selected shape panel changes, so successive scores are not repeated measurements on an identical set of clip/shape pairs. The standalone validation and test evaluations use panel index 110.

A.7 Split and experiment provenance

The split metadata is stored in `data/splits/hhi_stage2_v1/split_metadata.json`, with schema version 1, split seed 42, and split identifier

```
4c1c9d592e3dbeef15abe18cb50a3141820735849414ecc2d8959e9f38502263
```

It records the source-manifest hash, training/validation/test manifest and ID-list hashes, difficulty information, and the forced training reservation. The generator uses 100 difficulty strata. Its forced training reservation contains the 150 requested development identities and their existing mirrored partners, totalling 286 identities. This is a split-assignment count, distinct from the 150-clip development corpus used for the reported audit and design comparisons. Checkpoint/run split provenance refers to these artifacts, not just a verbal 90/5/5 ratio.

The documented HUMOS configuration selects `logs/humos/q6zbv2tu/checkpoints/latest-epoch=1599.ckpt`. The immutable HUMOS checkpoint checksum and the SMPLSim / asset-generation revision hashes are to be archived with the dataset release; a checkpoint path identifies the local workflow but is not a substitute for a checksum.

The principal implementation entry points are:

- Reference refinement: `tools/refine_humos_motion.py`; streamed per-clip processing: `tools/refine_humos_r2_per_clip.py`.

- Selected full-scale experiment: `examples/experiments/mimic/mlp_wide_stage2_discover_attention_slot_type.py`, inheriting the common attention experiment and enabling slot/type embeddings.
- Corrected development comparison: `tools/train_150_evalfix_ablation.sh` (cases A–E) and `tools/train_150_evalfix_unrefined.sh` (cases F–H), seed 0.
- Motion residency and evaluation: `protomotions/components/global_clip_pool.py` and `protomotions/agents/evaluators/mimic_evaluator.py`.
- Standalone split evaluation: `evaluate_validation_split.py` and `evaluate_test_split.py` under `protomotions/`. Each output records the checkpoint, split, reference version, and shape panel; the test evaluator additionally checks the test-manifest hash, count, and disjointness before running.

A.8 Unseen-body and repeated-seed evidence

The sole unseen-body study is the inside/outside replay of 9 October 2026. Its source report is `r2:proto-data/unseen_shapes_v2/eval/REPORT.md`; a local read-only snapshot is retained under `results/paper_evidence/unseen_shapes_v2/`, including both 128-row per-shape CSVs, comparison summaries, checkpoint identities, and rerun checks. The evaluator is `tools/analyze_unseen_morphology_generalization.py`; paired analysis uses `tools/compare_checkpoints_unseen_morphology.py`. The report records code revision `75324d9` at analysis time.

Frozen checkpoint SHA-256 values are:

```
refined (epoch 28,170):
4b83f20c93bf95fab389fc8935685ef8a2c7e46583f765e8c802421edb4479cb
fulldata (epoch 34,000):
7b57de8ec047c390c2881f8a630d7196a09da6dd08bfbae743f36d6542cd2192
```

Each checkpoint has its own frozen resolved inference configuration. All per-pair outputs and logs remain on R2; the local small-artifact snapshot is not a copy of the complete rollout records. The 233/23 clip partition, body manifests, and physical-property groups are defined in the source report.

Seed-1/2 architecture records are in the corresponding `results/hhi_150_evalfix_A_seed1` through `results/hhi_150_evalfix_D_seed2` run folders. Resumed logging segments must be joined by evaluation epoch before forming last-eight summaries. Across-seed standard deviation is computed from three run summaries, rather than from all scheduled evaluations pooled as independent observations.

For each final experiment, retain the resolved configuration, code revision, model/checkpoint checksum, body coefficient and asset manifests, split hashes, all resume segments, frame count, GPU model/count, and elapsed GPU-hours. For each evaluation, retain per-pair clip and asset IDs, panel selection, success/failure, valid horizon, and component errors in a local analysis file. The main full-scale W&B identifier is `3mvgad6f`; the corrected small-scale group is `hhi_150_evalfix_comparison`. The evaluation/sampling correction is recorded in commit `3685247`.

Immutable HUMOS-checkpoint and SMPLSim-revision checksums and public release of the retained validation/test per-motion records remain incomplete. The present study therefore does not claim a fully released benchmark.

B Development and Training Provenance

The tracking results use the original epoch-28,170 checkpoint; the unseen-body results also compare the frozen epoch-34,000 candidate. The monitoring snapshot below is a separate measurement at epoch 28,159. Historical single-seed comparisons and the full-scale run record are retained here to distinguish provenance from final performance.

B.1 Recorded full-scale validation result

Table 12 reports the final scheduled monitoring evaluation recorded for the selected full-scale slot/type run, at epoch 28,159 after approximately 11.07 billion simulated frames. The separately downloaded checkpoint is from epoch 28,170. These values are post-correction, morphology-matched evaluations, not measurements of that later checkpoint or of the clip-disjoint test split. Validation contains 1,048 held-out clips, each evaluated on one selected trained body, while the resident pool is difficulty-biased. Values are the stored distributed logging summaries of run `3mvgad6f` at epoch 28,159, and no confidence intervals or all-shape coverage are implied.

Table 12: Recorded full-scale monitoring snapshot.

Metric	Hard resident pool	Validation holdout
Success rate (%) ↑	89.84	98.47
Mean body-position error (m) ↓	0.0960	0.0698
Mean global body-rotation error (rad) ↓	0.2261	0.1760
Mean maximum-joint position error (m) ↓	0.1848	0.1412
Windowed normalized jerk ↓	1,442.8	1,151.7
High-jerk windows (%) ↓	31.05	27.25

The validation score is higher than the resident-pool score, while the reported errors are lower. This comparison is consistent with a curriculum that concentrates residency on difficult clips: the resident pool is not an independent uniform sample of the training distribution. Its 89.84% success rate should not be interpreted as the success rate of the entire training corpus, nor should the difference be described as validation outperforming an IID training sample.

Figure 5 shows the scheduled success rate over training for the full-scale slot/type run (`3mvgad6f`). The step near epoch 16.9k is the morphology-matched evaluation correction plus its sampling change, not an instantaneous learning event; pre-correction points use the earlier, sometimes mismatched reference-to-asset assignment and are not comparable to the post-correction curve. After the correction the holdout success rate climbs from about 0.96 to a stable 0.98–0.985. The resident-pool curve is noisier because residency is continually re-concentrated on hard clips.

The 98.47% result supports strong tracking of the selected held-out clip/body pairs under the specified threshold. It does not establish all-body success for those clips, unseen-body generalization, or an untouched test-set score. The thresholded success rate is complemented by position, rotation, and smoothness measures because a successful rollout can still have visible local errors or jitter.

The run record is available at the full-scale W&B run. The current table uses the recorded snapshot, not a newly executed evaluation.

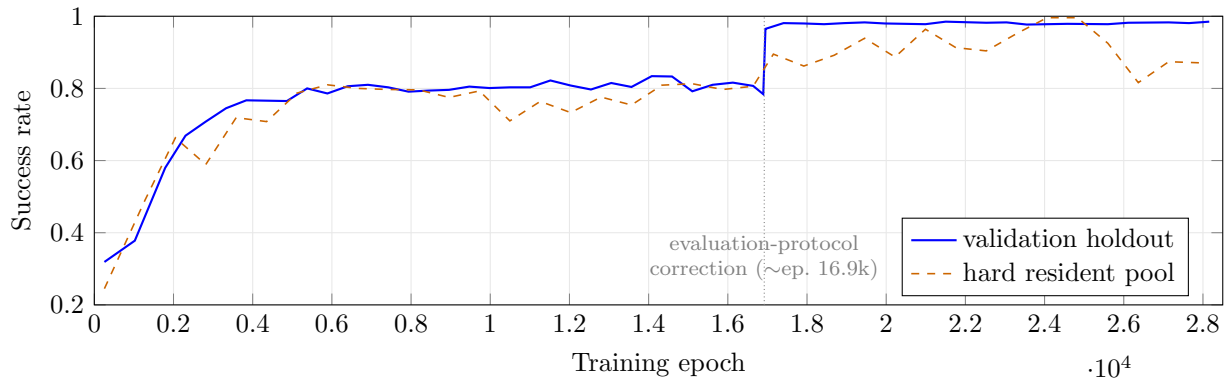


Figure 5: Scheduled success during full-scale training.

B.2 Historical single-seed reference grid

Table 13 lists the seed-0 development grid. Cases A–D use refined references and E–H unrefined references, so the two groups track different targets. Success shows the variation across eight evaluations within each run, and the other columns are their means; this within-run variation must not be interpreted as uncertainty across training runs. The main architecture conclusion uses the repeated refined runs of Table 7 instead. The columns gt err and gr err are the mean global body-position (m) and rotation (rad) errors, Jerk is the windowed normalized jerk, and HJ% is the percentage of high-jerk windows.

Table 13: Historical seed-0 development grid.

Case	Architecture / data	Success (%)	gt err	gr err	Jerk	HJ%
A	temporal MLP, refined	97.5 ± 0.8	0.109	0.207	1,729	33.3
B	attention, refined	97.5 ± 1.4	0.098	0.228	3,455	40.4
C	slot/type, refined	**98.1 ± 0.6**	0.097	0.226	**1,757**	34.2
D	slot/type + AdaLN, refined	97.1 ± 1.2	**0.090**	0.220	3,145	38.9
E	slot/type, unrefined	97.9 ± 0.6	0.102	0.219	1,422	29.6
F	temporal MLP, unrefined	96.2 ± 1.6	0.115	0.212	2,927	36.6
G	attention, unrefined	97.8 ± 1.2	0.098	0.219	3,117	34.9
H	slot/type + AdaLN, unrefined	97.5 ± 0.9	0.091	0.207	2,956	34.0

B.3 Full-scale run record

The full-scale MorphMimic run shares the architecture of case C exactly (same encoder, transformer, and head widths; `mlp_wide_stage2_discover_attention_slot_type.py` inherits these constants unchanged from the 150-clip experiment file), so its parameter count is also 51,238,582. Parameter counts for the four architectures (Table 7) are computed analytically from each run's resolved model configuration, because every linear layer is instantiated with `nn.LazyLinear` and has no fixed count outside a materialized model. The run used six A40 GPUs (`ddp`, 32-bit, 2,048 environments and an 8,192 minibatch per GPU) for roughly 28,170 epochs; the logged rank-0 wall time is about 77 h, on the order of 450 GPU-hours. An early restart after a memory-related failure resumed from the last checkpoint with the resident pool reduced from 256 to 128 clips per rank and the local cache multiplier set to 1.0; because configurations are reloaded from the pickle on resume, the frozen

Table 14: Outside coefficient tiers and simulated-mass bins, 233-clip panel.

Group	Bodies	Refined (%)	Full-data (%)
$\|\beta\|_\infty \in (3, 3.5]$	32	99.03	99.46
$\|\beta\|_\infty \in (3.5, 4]$	32	97.76	97.81
$\|\beta\|_\infty \in (4, 4.5]$	32	95.33	95.12
$\|\beta\|_\infty \in (4.5, 5]$	32	96.47	97.87
Mass $<$ 26.4 kg	2	100.00	100.00
Mass 26.4–144.4 kg	118	98.33	98.13
Mass $>$ 144.4 kg	8	79.02	88.62

configuration still records the pre-restart pool size of 256. The earlier failed launch was logged into the same run.

B.4 Reference refinement distributions

Figure 6 shows per-motion distributions (box: quartiles; whiskers: 5th–95th percentile) over the 19,200 development references. Stance-foot speed drops across almost the whole distribution, while position jerk rises across almost the whole distribution.

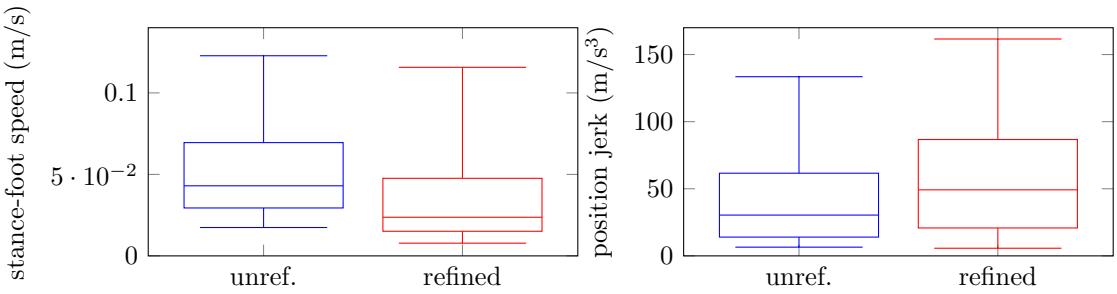


Figure 6: Reference sliding and jerk distributions before and after refinement.

C Additional Unseen-Body Results

This appendix adds the coefficient-tier and mass-bin summary and the checkpoint-regression list that support the unseen-body results. The complete per-body results for both panels, 256 rows in total, are in Online Resource 1.

C.1 Coefficient tiers and mass bins

Table 14 reports the predefined outside coefficient tiers and descriptive simulated-mass bins on the 233-clip panel. Tier and mass rows overlap, each partition covers all 128 bodies, and the mass thresholds are the trained-asset range. These are associations, not identified causes.

C.2 Checkpoint regressions

Table 15 lists all bodies whose paired success-change interval lies entirely below -2 percentage points on the 233-clip panel. These exploratory per-body intervals are not adjusted for multiple

Table 15: Bodies with a paired success decrease beyond 2 percentage points.

Panel	Asset	Refined (%)	Full-data (%)	Δ (pp)
Inside	female_e7bc8c92	92.3	79.0	−13.30
Inside	male_6d81a8f6	97.9	85.0	−12.88
Inside	male_ec3bb7f8	19.3	10.7	−8.58
Outside	female_edcf2449	94.8	58.4	−36.48
Outside	male_a7f6b877	65.7	40.8	−24.89
Outside	male_edcf2449	82.0	67.0	−15.02
Outside	female_e706da88	94.4	79.8	−14.59
Outside	female_bd0ec1ce	63.9	52.8	−11.16
Outside	female_f9bf3075	97.4	88.8	−8.58

Table 16: Cross-body control-demand variability, dynamic versus quasi-static clips.

Metric	Dynamic std [95% CI]	Quasi-static std [95% CI]
Normalized torque (frac. effort limit)	1.93 [1.71, 2.21] $\times 10^{-4}$	2.06 [1.86, 2.30] $\times 10^{-4}$
Mass-normalized power	0.292 [0.220, 0.370]	0.259 [0.210, 0.323]
Normalized jerk	350.2 [283.3, 416.9]	403.2 [303.3, 513.5]

comparisons.

D Exploratory Physical Analysis

As a secondary question, we ask whether cross-body variability in control demand differs between dynamic and quasi-static motions. The original MorphMimic checkpoint is replayed over the 150 development clips and all 128 trained bodies (19,200 pairs, zero recorded failed episodes under this analysis's own diagnostic). Keyword matching on the inherited captions assigns running, jumping, and kicking to a dynamic group (25 clips) and standing and sitting to a quasi-static group (37 clips); the other 88 clips are excluded, and the labels have not been manually validated. After subtracting each clip's across-body mean, Table 16 reports the standard deviation of the pooled residuals, with 95% intervals that resample clips as clusters (10,000 resamples). Torque, power, and jerk describe the simulated controller, not calibrated human biomechanics.

The spreads differ modestly and their separate intervals overlap, which is not a test of the class-level difference. A direct dynamic-minus-quasi-static contrast with clip-level resampling or permutation has not been estimated, so the analysis supports descriptive comparisons only, not a positive or negative interaction result.

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